```
# Install TensorFlow (if not already installed)
!pip install tensorflow

# Import libraries
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import matplotlib.pyplot as plt
import numpy as np

print("TensorFlow version:", tf.__version__)
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->te
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>
TensorFlow version: 2.19.0
```

```
# Load MNIST dataset
(X_train, y_train), (X_test, y_test) = keras.datasets.mnist.load_data()

# Normalize pixel values (0-255 → 0-1)
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten 28x28 images into 784-dimensional vectors
X_train = X_train.reshape(-1, 28*28)
X_test = X_test.reshape(-1, 28*28)

print("Training shape:", X_train.shape)
print("Test shape:", X_test.shape)
```

```
Training shape: (60000, 784)
Test shape: (10000, 784)
```

```
# Build Sequential ANN model
model = keras.Sequential([
    layers.Dense(128, activation='relu', input_shape=(784,)),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')  # 10 output classes (digits 0-9)
])

# Display model architecture
```

```
model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_3 (Dense)	(None, 128)	100,480
dense_4 (Dense)	(None, 64)	8,256
dense_5 (Dense)	(None, 10)	650

Total params: 109,386 (427.29 KB)
Trainable params: 109,386 (427.29 KB)

```
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

```
history = model.fit(
    X_train, y_train,
    epochs=10,
    batch_size=32,
    validation_split=0.2
)
```

```
Epoch 1/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.8646 - loss: 0.4632 - val_accuracy: 0.9616 - val_loss: 0.1318
Epoch 2/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.9661 - loss: 0.1136 - val_accuracy: 0.9645 - val_loss: 0.1146
Epoch 3/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.9776 - loss: 0.0734 - val_accuracy: 0.9694 - val_loss: 0.1010
Epoch 4/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 5s 4ms/step - accuracy: 0.9838 - loss: 0.0526 - val_accuracy: 0.9715 - val_loss: 0.1008
Epoch 5/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.9871 - loss: 0.0401 - val_accuracy: 0.9756 - val_loss: 0.0902
Epoch 6/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 5s 4ms/step - accuracy: 0.9901 - loss: 0.0312 - val_accuracy: 0.9728 - val_loss: 0.0970
Epoch 7/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 11s 4ms/step - accuracy: 0.9915 - loss: 0.0256 - val_accuracy: 0.9711 - val_loss: 0.1158
Epoch 8/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 5s 4ms/step - accuracy: 0.9924 - loss: 0.0240 - val_accuracy: 0.9746 - val_loss: 0.1027
Epoch 9/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.9936 - loss: 0.0188 - val_accuracy: 0.9738 - val_loss: 0.1035
Epoch 10/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 11s 5ms/step - accuracy: 0.9947 - loss: 0.0156 - val_accuracy: 0.9763 - val_loss: 0.0986
```

```
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", test_acc)
```

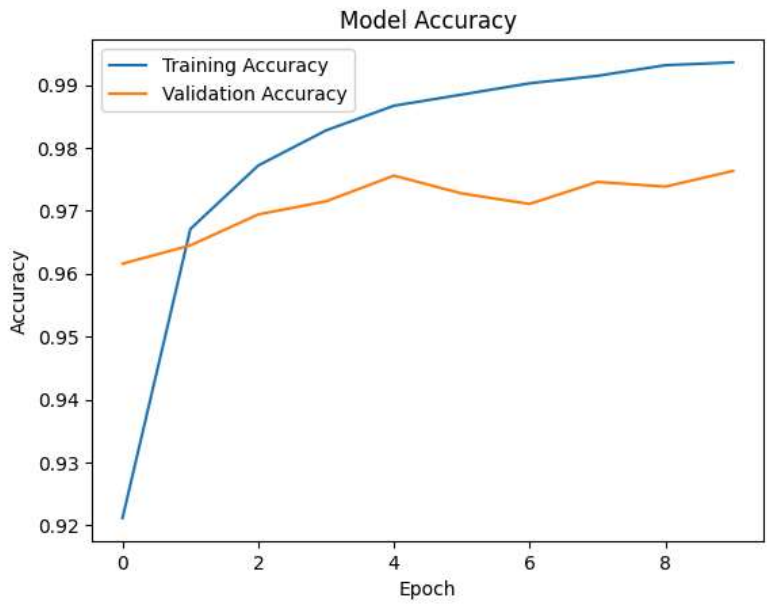
```
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step - accuracy: 0.9740 - loss: 0.1022
Test Accuracy: 0.9764999747276306
```

```
predictions = model.predict(X_test)

# Example: Show prediction for first test image
print("Predicted:", np.argmax(predictions[0]))
print("Actual:", y_test[0])
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 0s 1ms/step
Predicted: 7
Actual: 7
```

```
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.title("Model Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.show()
```


Model Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
0
2
4
6
8